CLASS 10 — MATHS · FORMULA SHEET

Chapter 14: **Probability** | Quick revision, one page

Classical (theoretical) probability

- $P(E) = \dfrac{\text{number of outcomes favourable to } E}{\text{total number of equally likely outcomes}}$.
- $0 \le P(E) \le 1$. Impossible event: $P = 0$. Sure (certain) event: $P = 1$.

Complementary events

- $P(E) + P(\overline{E}) = 1$, so $P(\overline{E}) = 1 - P(E)$ ($\overline{E}$ = 'not E').
- Sum of the probabilities of all elementary events of an experiment $= 1$.

Sample spaces

- One coin: H, T (2). Two coins: HH, HT, TH, TT (4). Three coins: 8 outcomes.
- One die: 1 to 6 (6). Two dice: $6 \times 6 = 36$ outcomes.

Two dice: useful counts

- Sum 2: 1, 3: 2, 4: 3, 5: 4, 6: 5, 7: 6, 8: 5, 9: 4, 10: 3, 11: 2, 12: 1 (total 36).
- Doublets: 6. Prime sums 2, 3, 5, 7, 11: 15 outcomes.

Counting tips

- Primes up to 20: 2, 3, 5, 7, 11, 13, 17, 19. Composite numbers have more than two factors; 1 is neither prime nor composite.
- 'A or B' when A and B overlap: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
- Adding objects to a bag changes both the favourable and the total counts.

MASTERMIND